2.The "Proof-of-Ordinance" Wildfire Watch

Mission Objective: Design a digital twin for a 3U CubeSat that detects thermal anomalies (wildfires) and immediately triggers a blockchain-based insurance payout for local farmers.

- **Web3 Element:** Use a **Smart Contract** as an "Insurance Oracle." If the simulation detects a heat signature in a specific GPS grid, the contract releases funds.
- **Constraint (Payload Choice):** High-resolution Thermal IR Imager (Simulation Model).
- **Constraint (Budget Estimation):** $2,500 for the digital twin software and "Gas Fees" for contract deployment.
- **Use Case:** Rapid, automated disaster relief without a middleman.

1. The Satellite and Digital Twin Concept

The mission utilizes a **3U CubeSat**, a standardized miniature satellite. In this context, the **Digital Twin** is a high-fidelity software replica of the satellite that runs on Earth. It processes telemetry and thermal data in a simulated environment to "double-check" the satellite's findings before the financial trigger is pulled.

- **Thermal IR Imager:** This is the "eye" of the mission. It captures Long-Wave Infrared (LWIR) data. Unlike standard cameras, it sees heat. It must be calibrated to ignore "noise" like hot asphalt or reflection from water.
- **The Power of 3U:** A 3U form factor (10x10x30 cm) is large enough to house the sophisticated optics and cooling systems required for infrared sensors, which often generate their own heat that can interfere with readings.

2. The Smart Contract as an "Oracle"

A **Smart Contract** is a self-executing piece of code on a blockchain. In this mission, the satellite acts as an **Oracle**—a bridge that provides real-world data to the digital contract.

- **Geofencing:** The contract contains a database of GPS coordinates belonging to insured farmers.
- **The Threshold Logic:** The contract waits for a specific data packet from the satellite. If the data says "Grid A-1 is 700°C," and the contract logic is programmed for "Payout if temp $> 500\text{°C}$," the funds are released instantly.
- **Eliminating the Middleman:** This prevents insurance companies from delaying payouts during a crisis, ensuring farmers have liquidity to buy feed, seed, or emergency supplies immediately.

3. Operational Challenges

While the concept is streamlined, the deployment faces several technical and economic hurdles:

- **Spatial Resolution:** A satellite at 500km altitude must have high-quality lenses to distinguish between a small campfire and a house-sized wildfire.

- **The $2,500 Budget:** This is the most difficult constraint. It suggests that the mission will use **"Leased Data"** or **Ground-Station-as-a-Service (GSaaS)** rather than building and launching a physical rocket from scratch. The budget likely covers cloud computing costs, blockchain "gas fees," and API access to existing thermal constellations.
- **False Positives:** The system must be smart enough to differentiate between a wildfire and a "controlled burn" or a large industrial furnace.

4. Mission Significance

This project represents a shift toward **Parametric Insurance**. Instead of paying for *actual damage* (which requires a human to visit and count dead trees), it pays based on a *parameter* (the heat signature).

- **Speed:** Payouts can happen in minutes, not months.
- **Objectivity:** Satellite data cannot be bribed or influenced; it is "Proof-of-Ordinance"—an immutable record of a physical event.

ROUND 1 SPECIFIC CONSTRAINTS:

1. Mission Objectives & Success Criteria

Every design must start with the **"Why."** In a hackathon, your mission objective isn't just a goal; it's a boundary.

- **Operational Lifetime:** How long must the satellite survive? (e.g., 6 months for a tech demo vs. 5 years for a weather satellite).
- **Target Orbit:** This dictates almost everything else.
    - **LEO (Low Earth Orbit):** Lower latency, higher resolution, but high atmospheric drag.
    - **GEO (Geostationary Orbit):** Constant coverage of one area, but extremely high launch costs and radiation.
- **Data Latency:** Does the data need to be "real-time" (for disaster relief) or can it be stored and forwarded once a day?

2. Payload Constraints

The payload is the "customer" of the satellite; everything else (power, thermal, comms) exists just to keep it happy.

- **SWaP-C:** This is the golden rule of satellite design: **S**ize, **W**eight, **a**nd **P**ower - **C**ost.
- **Mass & Volume:** Most hackathons use the "U" standard (1U = 10x10x10 cm). You'll likely be restricted to a 3U or 6U form factor.
- **Duty Cycle:** Can the payload run 24/7, or does it need to shut down to recharge batteries while the satellite is in the Earth's shadow?

3. Budget & Resource Estimation

In a real-world scenario (and a good hackathon), "unlimited money" isn't a thing.

- **Total Cost of Ownership (TCO):** You must account for:
    1. **Development:** Components and labor.
    2. **Launch:** Often calculated per kilogram (e.g., $50,000/kg for LEO).
    3. **Operations:** Ground station fees to talk to your satellite.
- **COTS vs. Custom:** Using **Commercial Off-The-Shelf (COTS)** parts is cheaper and faster but might not be "space-hardened" against radiation compared to custom, radiation-hardened components.

4. . Technical Constraints & Requirements

1. Payload & Sensing Constraints

- **Spatial Resolution (GSD):** The sensor must achieve a **Ground Sample Distance (GSD) of ≤ 100m**. This ensures the satellite can distinguish between an individual farm plot and general landscape heat.
- **Thermal Sensitivity (NETD):** The Noise Equivalent Temperature Difference must be **< 50mK**. This allows the sensor to distinguish subtle temperature gradients before a fire reaches peak intensity.
- **Spectral Bandwidth:** Targeting the **Mid-Wave Infrared (MWIR) at 3.7–5.0 µm**. This specific window is the "sweet spot" for detecting high-temperature anomalies (fire) while ignoring lower-temperature solar reflections.

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2. Data Integrity & Oracle Requirements

- **Hardware Root of Trust:** The satellite must use a **Trusted Execution Environment (TEE)** to digitally sign thermal data packets. This prevents "man-in-the-middle" attacks where a user might spoof fire data to trigger a fraudulent payout.
- **Binary Trigger Logic:** Data must be distilled into a **Boolean output (1 or 0)**.
    - *Logic:* IF (Temp > Threshold) AND (GPS == Registered_Grid) THEN Trigger_Payout.
- **Cloud Interference Mitigation:** The system requires an **Atmospheric Correction Algorithm** within the Digital Twin to ensure that smoke or heavy cloud cover doesn't lead to "False Negatives" (failing to see a fire that is actually there).

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3. Software & Processing Requirements

- **Edge Processing (On-Orbit):** To save on downlink costs, the 3U CubeSat must perform **on-board thresholding**. Instead of sending a high-resolution 50MB image, it sends a tiny **kilobyte-sized metadata packet** confirming the anomaly.
- **Revisit Frequency:** Since one satellite has limited coverage, the system must utilize **Simplified General Perturbations (SGP4) propagators** to predict exactly when the satellite will be over the insured grid to "arm" the smart contract.

- **Digital Twin Synchronization:** The virtual model must sync with the physical satellite every **orbit cycle (~90 mins)** to update health status and sensor calibration offsets.

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4. Financial & Deployment Constraints ($2,500 Budget)

- **Network Selection:** Use of a **Layer 2 (L2) Blockchain** (e.g., Polygon or Arbitrum). This is a hard requirement to keep transaction/gas fees under **$0.01 per payout**, preserving the limited budget.
- **Infrastructure:** Utilization of **Ground-Station-as-a-Service (GSaaS)**. By leasing antenna time only when the satellite passes over, the mission avoids the massive capital cost of building a private ground station.
- **Tech Stack:** * **Language:** Python (using Skyfield and OpenCV).
    - **Smart Contracts:** Solidity (OpenZeppelin standards for security).
    - **Oracles:** Chainlink (to bridge off-chain satellite data to the on-chain contract).